

31 - Halogen compounds

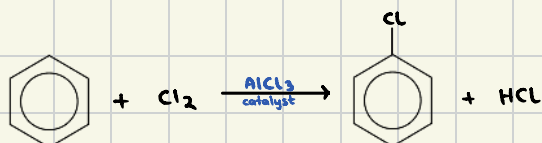
Production of halogen compounds

→ chlorination of benzene

reagent : chlorine (Cl_2)

catalyst : AlCl_3 (or Fe)

conditions : anhydrous conditions & room temp.

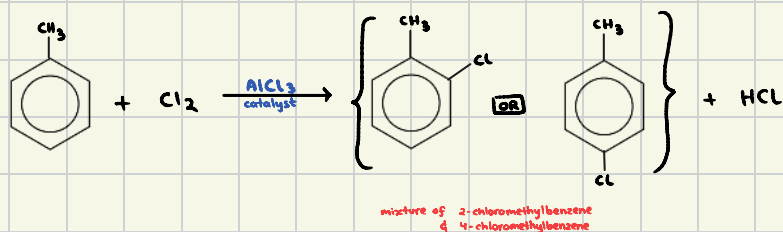


→ chlorination of methylbenzene

reagent : chlorine (Cl_2)

catalyst : AlCl_3 (or Fe)

conditions : anhydrous conditions & room temp.

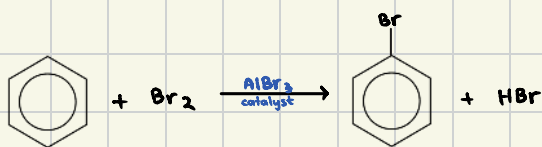


→ bromination of benzene

reagent : bromine (Br_2)

catalyst : AlBr_3 (or Fe)

conditions : anhydrous conditions & room temp.

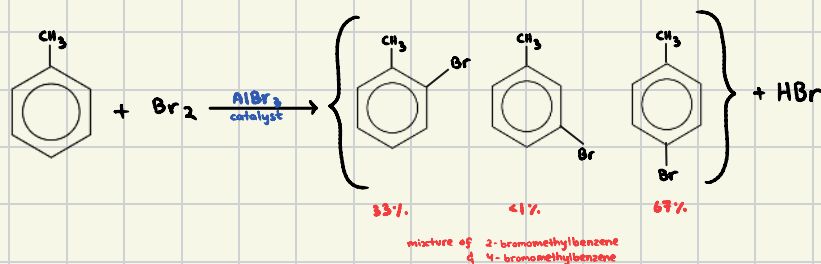


→ bromination of methylbenzene

reagent : chlorine (Cl_2)

catalyst : AlCl_3 (or Fe)

conditions : anhydrous conditions & room temp.

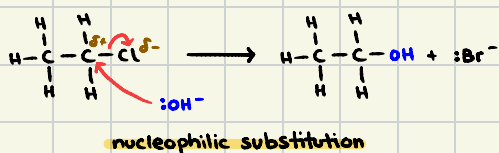


Halogenoalkane VS Halogenoarene

Halogenoarenes are VERY unreactive compared to halogenoalkanes

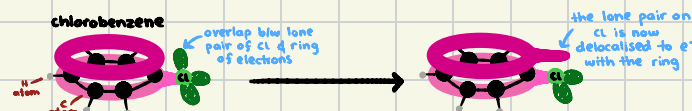
WHY?

Halogenoalkane
more reactive



- 1) nucleophile attacks the slightly +ve C atom
- 2) covalent bond is formed b/w C-OH, & C-Cl breaks
- 3) overall, the halogen is replaced by the nucleophile

Halogenoarene
less reactive



→ halogenoarenes do NOT undergo nucleophilic substitution unless you introduce harsh conditions, such as 200°C & 200 atm

why? → the C-Cl bond is very strong & cannot be broken easily

→ one of the lone pair of e^- on the Cl will interact with the π system of the ring

→ this causes the C-Cl bond to have a partial double-bond character, which strengthens the bond